University of Texas at Austin
McCombs School of Business
INVESTMENT THEORY/ACF
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Project I: Creating Custom Equity Indexes

This assignment is designed for MSBA/Finance students learning empirical methods in asset pricing, portfolio analysis, and applied financial data science. Students will work with **Python, WRDS, and CRSP data** to design, implement, and analyze custom equity indexes, comparing their construction to industry benchmarks.

Objectives

By completing this project, students will:

- Gain experience retrieving, cleaning, and structuring **financial datasets** from WRDS/CRSP.
- Understand and implement **index weighting methodologies** (equal-, value-, and price-weighted).
- Apply **reconstitution and rebalancing rules** while avoiding look-ahead bias.
- Incorporate **delisting returns** to mitigate survivorship bias.
- Benchmark custom indexes against **major ETFs** (SPY, IWM, QQQ).
- Visualize performance and calculate **correlation matrices** of log returns.
- Interpret results through the lens of diversification, bias mitigation, and empirical finance best practices.

Assignment Tasks

Part I. Data Preparation

1. Connect to WRDS via Python (wrds library).
2. Query **CRSP Monthly Stock File (MSF)** and join with name history tables to filter:
   - U.S. common shares only: shrcd $\in \{10,11\}$
   - Major exchanges only: exchcd $\in \{1,2,3\}$
3. Clean the data:
   - Standardize price signs (absolute value).
   - Convert shares outstanding from thousands.
   - Compute **market capitalization** for each security and month.
4. Incorporate **delisting returns** (msedelist) to compute effective monthly returns[1]

Part II. Index Construction

1. Implement **monthly reconstitution and rebalancing**:
   - Use **t−1 information only** to avoid look-ahead bias.
   - Select the **Top-N securities by market cap** at each month-end.
   - Carry forward index values in months with incomplete data.

[1]One method is: $r_{eff} = (1 + r_{CRSP}) \times (1 + r_{delist}) - 1$, where r_{CRSP} is the return from the CRSP data and r_{delist} is the delisting return for the security.

2. Build three indexes:
    - **Equal-Weighted (EW)**
    - **Value-Weighted (VW)**
    - **Price-Weighted (PW)**
3. Initialize all indexes with base level = 100.

Part III. Benchmarking & Analytics

1. Retrieve benchmark ETF series (SPY, IWM, QQQ) from CRSP.
2. Compute **log returns** for custom indexes and ETFs.
3. Generate:
    - **Time-series plots** of index levels (custom vs. ETFs).
    - **Heatmap correlation matrix** of monthly log returns.
4. Interpret:
    - Co-movement of custom indexes with ETFs.
    - Effects of weighting schemes on return/volatility.
    - Bias mitigation (look-ahead, survivorship).

Part IV. Deliverables

Each group must submit:

1. **Jupyter Notebook** (.ipynb) containing:
    - Modular, well-commented functions for each step (data, returns, index build, benchmarks, plots).
    - Code structured for readability and reproducibility.
2. **Written Report** (.pdf) (3-5 pages, not including exhibits or illustrations) including:
    - Overview of methodology.
    - Key implementation choices (e.g., Top-N threshold, reconstitution logic).
    - Plots and correlation matrix results.
    - Discussion of findings: performance differences, diversification, robustness.
    - Reflection on challenges (data quality, missing values, survivorship).

APPENDIX: AI Usage Policy for Project Assignment

1. **Permitted Uses**
   - Students may use Large Language Models (LLMs) such as ChatGPT, Claude, or Copilot to assist with programming tasks only.
   - Acceptable uses include:
     - Debugging code and troubleshooting errors.
     - Asking for clarification of Python syntax, library functions, or error messages.
     - Generating short code snippets that you then review, adapt, and integrate into your own work.

2. **Prohibited Uses**
   - Students may not use any AI tool to draft or write any portion of their written report/memo. The written report must be entirely student-authored.
   - Wholesale outsourcing of coding or analysis to AI tools (e.g., pasting the assignment and asking the LLM to generate the entire solution) is strictly prohibited.
   - Students must not rely uncritically on AI outputs. Any AI-generated code must be reviewed, tested, and validated by the student.

3. **Citation & Documentation Requirement**
   - Any use of AI tools must be transparently disclosed in the appendix to your report. This disclosure must include:
     - The tool/model name (e.g., ChatGPT GPT-4, Claude 3.5).
     - The date of usage.
     - The exact prompts provided to the AI.
     - The verbatim responses received.
   - All AI-related materials should be placed in an Exhibit (Appendix) to your report and clearly labeled.

4. **Responsibility and Verification**
   - You are fully responsible for the accuracy, validity, and originality of any code you submit, regardless of whether AI tools were consulted.
   - Treat AI outputs as *unverified suggestions* — they often contain errors, hallucinations, or incomplete logic. Verification and correctness remain the student's responsibility.
   - Failure to disclose AI usage will be treated as an academic integrity violation.

You may use AI tools as a *coding assistant* but not as a *report writer.* Any use must be disclosed with full transparency. AI should support your learning, not replace it!